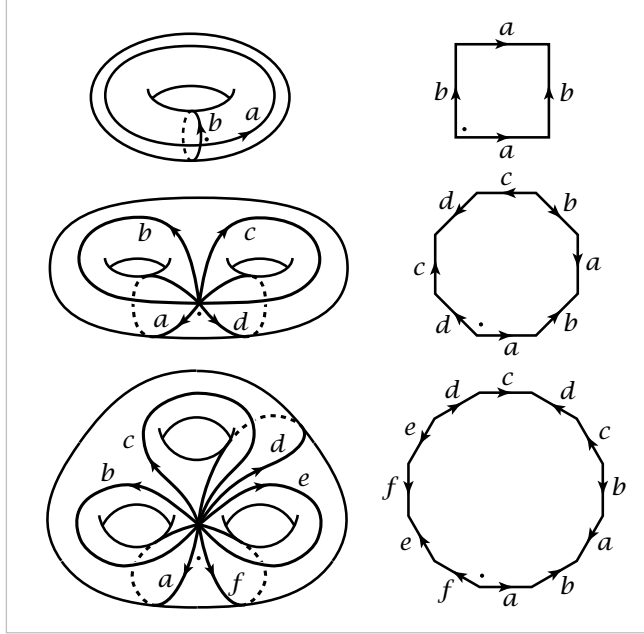


Cell Complexes

A familiar way of constructing the torus $S^1 \times S^1$ is by identifying opposite sides of a square. More generally, an orientable surface M_g of genus g can be constructed from a polygon with $4g$ sides by identifying pairs of edges, as shown in the figure in the first three cases $g = 1, 2, 3$. The $4g$ edges of the polygon become a union of $2g$ circles in the surface, all intersecting in a single point. The interior of the polygon can be thought of as an open disk, or a **2-cell**, attached to the union of the $2g$ circles. One can also regard the union of the circles as being obtained from their common point of intersection, by attaching $2g$ open arcs, or **1-cells**. Thus the surface can be built up in stages: Start with a point, attach 1-cells to this point, then attach a 2-cell.



A natural generalization of this is to construct a space by the following procedure:

- (1) Start with a discrete set X^0 , whose points are regarded as 0-cells.
- (2) Inductively, form the **n -skeleton** X^n from X^{n-1} by attaching n -cells e_α^n via maps $\varphi_\alpha: S^{n-1} \rightarrow X^{n-1}$. This means that X^n is the quotient space of the disjoint union $X^{n-1} \coprod_\alpha D_\alpha^n$ of X^{n-1} with a collection of n -disks D_α^n under the identifications $x \sim \varphi_\alpha(x)$ for $x \in \partial D_\alpha^n$. Thus as a set, $X^n = X^{n-1} \coprod_\alpha e_\alpha^n$ where each e_α^n is an open n -disk.
- (3) One can either stop this inductive process at a finite stage, setting $X = X^n$ for some $n < \infty$, or one can continue indefinitely, setting $X = \bigcup_n X^n$. In the latter case X is given the weak topology: A set $A \subset X$ is open (or closed) iff $A \cap X^n$ is open (or closed) in X^n for each n .

A space X constructed in this way is called a **cell complex** or **CW complex**. The explanation of the letters ‘CW’ is given in the Appendix, where a number of basic topological properties of cell complexes are proved. The reader who wonders about various point-set topological questions lurking in the background of the following discussion should consult the Appendix for details.

If $X = X^n$ for some n , then X is said to be finite-dimensional, and the smallest such n is the **dimension** of X , the maximum dimension of cells of X .

Example 0.1. A 1-dimensional cell complex $X = X^1$ is what is called a **graph** in algebraic topology. It consists of vertices (the 0-cells) to which edges (the 1-cells) are attached. The two ends of an edge can be attached to the same vertex.

Example 0.2. The house with two rooms, pictured earlier, has a visually obvious 2-dimensional cell complex structure. The 0-cells are the vertices where three or more of the depicted edges meet, and the 1-cells are the interiors of the edges connecting these vertices. This gives the 1-skeleton X^1 , and the 2-cells are the components of the remainder of the space, $X - X^1$. If one counts up, one finds there are 29 0-cells, 51 1-cells, and 23 2-cells, with the alternating sum $29 - 51 + 23$ equal to 1. This is the **Euler characteristic**, which for a cell complex with finitely many cells is defined to be the number of even-dimensional cells minus the number of odd-dimensional cells. As we shall show in Theorem 2.44, the Euler characteristic of a cell complex depends only on its homotopy type, so the fact that the house with two rooms has the homotopy type of a point implies that its Euler characteristic must be 1, no matter how it is represented as a cell complex.

Example 0.3. The sphere S^n has the structure of a cell complex with just two cells, e^0 and e^n , the n -cell being attached by the constant map $S^{n-1} \rightarrow e^0$. This is equivalent to regarding S^n as the quotient space $D^n / \partial D^n$.

Example 0.4. Real projective n -space $\mathbb{R}P^n$ is defined to be the space of all lines through the origin in $\mathbb{R}^{n+1}$. Each such line is determined by a nonzero vector in $\mathbb{R}^{n+1}$, unique up to scalar multiplication, and $\mathbb{R}P^n$ is topologized as the quotient space of $\mathbb{R}^{n+1} - \{0\}$ under the equivalence relation $v \sim \lambda v$ for scalars $\lambda \neq 0$. We can restrict to vectors of length 1, so $\mathbb{R}P^n$ is also the quotient space $S^n / (v \sim -v)$, the sphere with antipodal points identified. This is equivalent to saying that $\mathbb{R}P^n$ is the quotient space of a hemisphere D^n with antipodal points of ∂D^n identified. Since ∂D^n with antipodal points identified is just $\mathbb{R}P^{n-1}$, we see that $\mathbb{R}P^n$ is obtained from $\mathbb{R}P^{n-1}$ by attaching an n -cell, with the quotient projection $S^{n-1} \rightarrow \mathbb{R}P^{n-1}$ as the attaching map. It follows by induction on n that $\mathbb{R}P^n$ has a cell complex structure $e^0 \cup e^1 \cup \dots \cup e^n$ with one cell e^i in each dimension $i \leq n$.

Example 0.5. Since $\mathbb{R}P^n$ is obtained from $\mathbb{R}P^{n-1}$ by attaching an n -cell, the infinite union $\mathbb{R}P^\infty = \bigcup_n \mathbb{R}P^n$ becomes a cell complex with one cell in each dimension. We can view $\mathbb{R}P^\infty$ as the space of lines through the origin in $\mathbb{R}^\infty = \bigcup_n \mathbb{R}^n$.

Example 0.6. Complex projective n -space $\mathbb{C}P^n$ is the space of complex lines through the origin in $\mathbb{C}^{n+1}$, that is, 1-dimensional vector subspaces of $\mathbb{C}^{n+1}$. As in the case of $\mathbb{R}P^n$, each line is determined by a nonzero vector in $\mathbb{C}^{n+1}$, unique up to scalar multiplication, and $\mathbb{C}P^n$ is topologized as the quotient space of $\mathbb{C}^{n+1} - \{0\}$ under the

equivalence relation $v \sim \lambda v$ for $\lambda \neq 0$. Equivalently, this is the quotient of the unit sphere $S^{2n+1} \subset \mathbb{C}^{n+1}$ with $v \sim \lambda v$ for $|\lambda| = 1$. It is also possible to obtain $\mathbb{CP}^n$ as a quotient space of the disk D^{2n} under the identifications $v \sim \lambda v$ for $v \in \partial D^{2n}$, in the following way. The vectors in $S^{2n+1} \subset \mathbb{C}^{n+1}$ with last coordinate real and nonnegative are precisely the vectors of the form $(w, \sqrt{1 - |w|^2}) \in \mathbb{C}^n \times \mathbb{C}$ with $|w| \leq 1$. Such vectors form the graph of the function $w \mapsto \sqrt{1 - |w|^2}$. This is a disk D_+^{2n} bounded by the sphere $S^{2n-1} \subset S^{2n+1}$ consisting of vectors $(w, 0) \in \mathbb{C}^n \times \mathbb{C}$ with $|w| = 1$. Each vector in S^{2n+1} is equivalent under the identifications $v \sim \lambda v$ to a vector in D_+^{2n} , and the latter vector is unique if its last coordinate is nonzero. If the last coordinate is zero, we have just the identifications $v \sim \lambda v$ for $v \in S^{2n-1}$.

From this description of $\mathbb{CP}^n$ as the quotient of D_+^{2n} under the identifications $v \sim \lambda v$ for $v \in S^{2n-1}$ it follows that $\mathbb{CP}^n$ is obtained from $\mathbb{CP}^{n-1}$ by attaching a cell e^{2n} via the quotient map $S^{2n-1} \rightarrow \mathbb{CP}^{n-1}$. So by induction on n we obtain a cell structure $\mathbb{CP}^n = e^0 \cup e^2 \cup \dots \cup e^{2n}$ with cells only in even dimensions. Similarly, $\mathbb{CP}^\infty$ has a cell structure with one cell in each even dimension.

After these examples we return now to general theory. Each cell e_α^n in a cell complex X has a **characteristic map** $\Phi_\alpha: D_\alpha^n \rightarrow X$ which extends the attaching map φ_α and is a homeomorphism from the interior of D_α^n onto e_α^n . Namely, we can take Φ_α to be the composition $D_\alpha^n \hookrightarrow X^{n-1} \amalg_\alpha D_\alpha^n \rightarrow X^n \hookrightarrow X$ where the middle map is the quotient map defining X^n . For example, in the canonical cell structure on S^n described in Example 0.3, a characteristic map for the n -cell is the quotient map $D^n \rightarrow S^n$ collapsing ∂D^n to a point. For $\mathbb{RP}^n$ a characteristic map for the cell e^i is the quotient map $D^i \rightarrow \mathbb{RP}^i \subset \mathbb{RP}^n$ identifying antipodal points of ∂D^i , and similarly for $\mathbb{CP}^n$.

A **subcomplex** of a cell complex X is a closed subspace $A \subset X$ that is a union of cells of X . Since A is closed, the characteristic map of each cell in A has image contained in A , and in particular the image of the attaching map of each cell in A is contained in A , so A is a cell complex in its own right. A pair (X, A) consisting of a cell complex X and a subcomplex A will be called a **CW pair**.

For example, each skeleton X^n of a cell complex X is a subcomplex. Particular cases of this are the subcomplexes $\mathbb{RP}^k \subset \mathbb{RP}^n$ and $\mathbb{CP}^k \subset \mathbb{CP}^n$ for $k \leq n$. These are in fact the only subcomplexes of $\mathbb{RP}^n$ and $\mathbb{CP}^n$.

There are natural inclusions $S^0 \subset S^1 \subset \dots \subset S^n$, but these subspheres are not subcomplexes of S^n in its usual cell structure with just two cells. However, we can give S^n a different cell structure in which each of the subspheres S^k is a subcomplex, by regarding each S^k as being obtained inductively from the equatorial S^{k-1} by attaching two k -cells, the components of $S^k - S^{k-1}$. The infinite-dimensional sphere $S^\infty = \bigcup_n S^n$ then becomes a cell complex as well. Note that the two-to-one quotient map $S^\infty \rightarrow \mathbb{RP}^\infty$ that identifies antipodal points of S^∞ identifies the two n -cells of S^∞ to the single n -cell of $\mathbb{RP}^\infty$.

In the examples of cell complexes given so far, the closure of each cell is a subcomplex, and more generally the closure of any collection of cells is a subcomplex. Most naturally arising cell structures have this property, but it need not hold in general. For example, if we start with S^1 with its minimal cell structure and attach to this a 2-cell by a map $S^1 \rightarrow S^1$ whose image is a nontrivial subarc of S^1 , then the closure of the 2-cell is not a subcomplex since it contains only a part of the 1-cell.

Operations on Spaces

Cell complexes have a very nice mixture of rigidity and flexibility, with enough rigidity to allow many arguments to proceed in a combinatorial cell-by-cell fashion and enough flexibility to allow many natural constructions to be performed on them. Here are some of those constructions.

Products. If X and Y are cell complexes, then $X \times Y$ has the structure of a cell complex with cells the products $e_\alpha^m \times e_\beta^n$ where e_α^m ranges over the cells of X and e_β^n ranges over the cells of Y . For example, the cell structure on the torus $S^1 \times S^1$ described at the beginning of this section is obtained in this way from the standard cell structure on S^1 . For completely general CW complexes X and Y there is one small complication: The topology on $X \times Y$ as a cell complex is sometimes finer than the product topology, with more open sets than the product topology has, though the two topologies coincide if either X or Y has only finitely many cells, or if both X and Y have countably many cells. This is explained in the Appendix. In practice this subtle issue of point-set topology rarely causes problems, however.

Quotients. If (X, A) is a CW pair consisting of a cell complex X and a subcomplex A , then the quotient space X/A inherits a natural cell complex structure from X . The cells of X/A are the cells of $X - A$ plus one new 0-cell, the image of A in X/A . For a cell e_α^n of $X - A$ attached by $\varphi_\alpha: S^{n-1} \rightarrow X^{n-1}$, the attaching map for the corresponding cell in X/A is the composition $S^{n-1} \rightarrow X^{n-1} \rightarrow X^{n-1}/A^{n-1}$.

For example, if we give S^{n-1} any cell structure and build D^n from S^{n-1} by attaching an n -cell, then the quotient D^n/S^{n-1} is S^n with its usual cell structure. As another example, take X to be a closed orientable surface with the cell structure described at the beginning of this section, with a single 2-cell, and let A be the complement of this 2-cell, the 1-skeleton of X . Then X/A has a cell structure consisting of a 0-cell with a 2-cell attached, and there is only one way to attach a cell to a 0-cell, by the constant map, so X/A is S^2 .

Suspension. For a space X , the **suspension** SX is the quotient of $X \times I$ obtained by collapsing $X \times \{0\}$ to one point and $X \times \{1\}$ to another point. The motivating example is $X = S^n$, when $SX = S^{n+1}$ with the two ‘suspension points’ at the north and south poles of S^{n+1} , the points $(0, \dots, 0, \pm 1)$. One can regard SX as a double cone

